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MODERN INSTITUTE OF TECHNOLOGY AND RESEARCH CENTRE, ALWAR

NAME OF FACULTY: RAJESH KUMAR

SUB & CODE: Database Management Systems (4CS4-04, 4AD4-04, 4AM4-04)

SEMESTER: IV SESSION: 2023-24 (Even SEM.)

BRANCH: CS & AI/DS BATCH: A & B

UNIT: II SUBMISSON: 23-2-2024

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References:

1. Database System Concepts 6th Edition Abraham Silberschatz, Henry F Korth
2. Fundamentals of Database Systems by Elmsari, Navathe, 5th Edition Pearson

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UNIT-1 (Overview and History of DBMS)
LECTURE NO: -02

What is Data?

Data is a collection of facts and figures. The data collection was increasing day to day and they needed to be stored in a device or a software.

Data becomes information when it is processed, turning it into something meaningful.

What is a Database?

A database is a collection of related data which represents some aspect of the real world. A database system is designed to be built and populated with data for a certain task.

A DBMS is a software that allows creation, definition and manipulation of database, allowing users to store, process and analyse data easily. DBMS provides us with an interface or a tool, to perform various operations like creating database, storing data in it, updating data, creating tables in the database and a lot more.

Example of a DBMS:-

The STUDENT file stores data of each student

The COURSE file stores contain data on each course.

The SECTION stores the information about sections in a particular course.

History of Database System (DBMS)

1950's and early 1960's

In 1950s magnetic tape was introduced for storage of data

The data storage process consists of reading data from many tapes and writing of data to a new tape. For input, process punched cards used, and for output, the printer was used.

1970's-72's

In this year two main relational database system was created, which was developed at UBC, and system R. E.F Codd proposed the relational model for the database in a landmark paper on how to think about database. This gave a new concept about the database.

1976

P-Chen introduces a new database model known as ER Entity-Relationship. This design is basically made to focus on data applications. It gave another important insight into the conceptual data model. Such a higher-level melding allows the designers to focus on the data instead of a logical structure.

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Early 1980's

SQL (Structured Query Language) became standard. IBM launched DB2.the development of the IBM PC gave rise to many DB companies and products such PARADOX, RBASE, Database Management System also introduce in this period. The relational database system became a success as rapid increase in computer sales. This increased the popularity of the network and hierarchical database models.

Early 1990's

In early 1990s much work was done on client tools for application as development such as PowerBuilder (Microsoft), Oracle Developer. The client-server model for computing became the norm for future business decisions. The work on the object database system also started in this year.

Mid 1990's

The usable internet /WWW was introduced in the middle of this year.it allows remote access to the computer systems. The concept of Web/DB started to grow. The advancement in the database system was started. Average desktops users began to use the client-server database.

Late 1990's

In the late 1990s, the online business was increased resulting in a rise in demand for internet database connectors like Active server page, Java Servlets, FrontPage, Dream Weaver. Enterprise Java Beans, Oracle Developer 2000 etc.in this year internet companies worked for Web/Internet connectors.

Early 21st Century

The growth of DB applications continued in the early 21st century. The main companies that predominated the large DB market are IBM. Microsoft and oracle.

Today

Today's database is used anywhere in our day to day life. In the present day, there are many new players in the non-relational database space offering particular solutions. Many powerful technologies are introducing in the database. The database is very helpful in every field to keep records save.

Characteristics of Database Management System

A database management system has following characteristics:

1. **Data stored into Tables**: Data is never directly stored into the database. Data is stored into tables, created inside the database. DBMS also allows to have relationships between tables which makes the data more meaningful and connected. You can easily understand what type of data is stored where by looking at all the tables created in a database.

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2. **Reduced Redundancy:** In the modern world hard drives are very cheap, but earlier when hard drives were too expensive, unnecessary repetition of data in database was a big problem. But DBMS follows Normalisation which divides the data in such a way that repetition is minimum.

3. **Data Consistency:** On Live data, i.e. data that is being continuously updated and added, maintaining the consistency of data can become a challenge. But DBMS handles it all by itself.

4. **Support Multiple user and Concurrent Access:** DBMS allows multiple users to work on it (update, insert, delete data) at the same time and still manages to maintain the data consistency.

5. **Query Language:** DBMS provides users with a simple Query language, using which data can be easily fetched, inserted, deleted and updated in a database.

6. **Security:** The DBMS also takes care of the security of data, protecting the data from un-authorised access. In a typical DBMS, we can create user accounts with different access permissions, using which we can easily secure our data by restricting user access.

7. **DBMS supports transactions**, which allows us to better handle and manage data integrity in real world applications where multi-threading is extensively used.

Application of DBMS: -

Banking	For customer information, account activities, payments, deposits, loans, etc.
Airlines	For reservations and schedule information.
Universities	For student information, course registrations, colleges and grades.
Telecommunication	It helps to keep call records, monthly bills, maintaining balances, etc.
Finance	For storing information about stock, sales, and purchases of financial instruments like stocks and bonds.

Important Point: -

- DBMS definition: A database is a collection of related data which represents some aspect of the real world
- The full form of DBMS is Database Management System. DBMS stands for Database Management System is a software for storing and retrieving users' data by considering appropriate security measures.
- DBMS Provides security and removes redundancy.

References: -

1. Database System Concepts 6th Edition Abraham Silberschatz, Henry F. Korth

2.Fundamentals of Database Systems" by Elmsari, Navathe, 5th Edition, Pearson Education (2008).

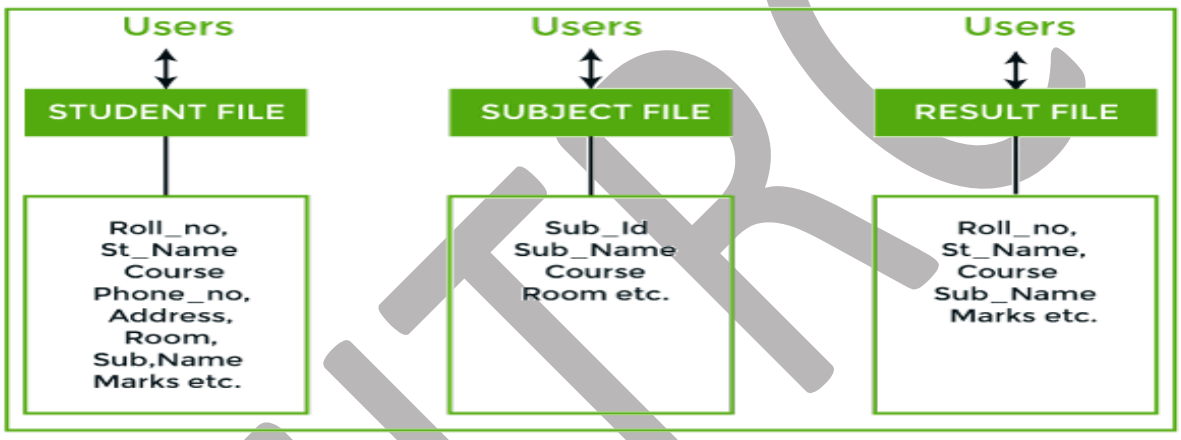
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UNIT-1 (File System v/s DBMS)
LECTURE NO: -03

What is a File system?

A file system is a technique of arranging the files in a storage medium like a hard disk, pen drive, DVD, etc. It helps you to organizes the data and allows easy retrieval of files when they are required. It mostly consists of different types of files like mp3, mp4, txt, doc, etc. that are grouped into directories.

A file system enables you to handle the way of reading and writing data to the storage medium. It is directly installed into the computer with the Operating systems such as Windows and Linux.



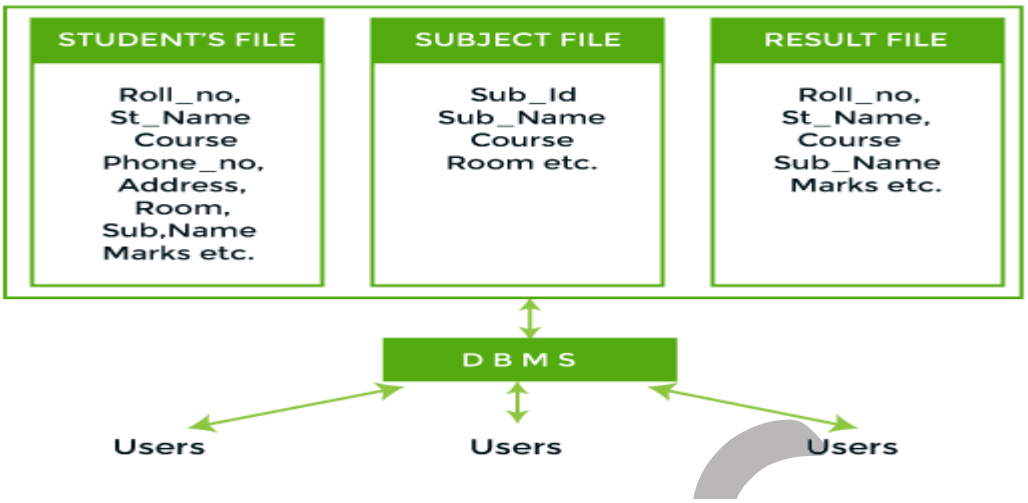
In the above figure:

Consider an example of a student's file system. The student file will contain information regarding the student (i.e. roll no, student name, course etc.). Similarly, we have a subject file that contains information about the subject and the result file which contains the information regarding the result.

What is DBMS?

Database Management System (DBMS) is a software for storing and retrieving user's data while considering appropriate security measures. It consists of a group of programs that manipulate the database. The DBMS accepts the request for data from an application and instructs the DBMS engine to provide the specific data. In large systems, a DBMS helps users and other third-party software to store and retrieve data.

File System v/s DBMS



In the above figure,

In the above figure, duplication of data is reduced due to centralization of data.

There are the following differences between DBMS and File systems:

Basis	DBMS Approach	File System Approach
Meaning	DBMS is a collection of data. In DBMS, the user is not required to write the procedures.	The file system is a collection of data. In this system, the user has to write the procedures for managing the database.
Sharing of data	Due to the centralized approach, data sharing is easy.	Data is distributed in many files, and it may be of different formats, so it isn't easy to share data.
Data Abstraction	DBMS gives an abstract view of data that hides the details.	The file system provides the detail of the data representation and storage of data.
Security and Protection	DBMS provides a good protection mechanism.	It isn't easy to protect a file under the file system.
Recovery Mechanism	DBMS provides a crash recovery mechanism, i.e., DBMS protects the user from system failure.	The file system doesn't have a crash mechanism, i.e., if the system crashes while entering some data, then the content of the file will be lost.
Manipulation Techniques	DBMS contains a wide variety of sophisticated techniques to store and retrieve the data.	The file system can't efficiently store and retrieve the data.

File System v/s DBMS

Concurrency Problems	DBMS takes care of Concurrent access of data using some form of locking.	In the File system, concurrent access has many problems like redirecting the file while deleting some information or updating some information.
Where to use	Database approach used in large systems which interrelate many files.	File system approach used in large systems which interrelate many files.
Cost	The database system is expensive to design.	The file system approach is cheaper to design.

Advantages of File system

- Enforcement of development and maintenance standards.
- Helps you to reduce redundancy
- Avoid inconsistency across file maintenance to get the integrity of data independence.
- It is more efficient and cost less than a DBMS in certain situations.
- The design of file processing is simpler than designing Database.

Disadvantages of File system

- Each application has its data file so, the same data may have to be recorded and stored many times.
- Data dependence in the file processing system are data-dependent, but, the problem is incompatible with file format.
- Limited data sharing.
- The problem with security.
- Time-consuming.

Advantages of DBMS system

- DBMS offers a variety of techniques to store & retrieve data
- Uniform administration procedures for data
- Application programmers never exposed to details of data representation and Storage.
- A DBMS uses various powerful functions to store and retrieve data efficiently.
- The DBMS implies integrity constraints to get a high level of protection against prohibited access to data.
- Reduced Application Development Time
- Reduction of redundancy

Disadvantages of the DBMS system

- Cost of Hardware and Software of a DBMS is quite high, which increases the budget of your organization.
- Most database management systems are often complex systems, so the training for users to use the DBMS is required.
- The use of the same program at a time by many users sometimes lead to the loss of some data.
- DBMS can't perform sophisticated calculations
- Data-sets begins to grow large as it provides a more predictable query response time.

Storage System in DBMS

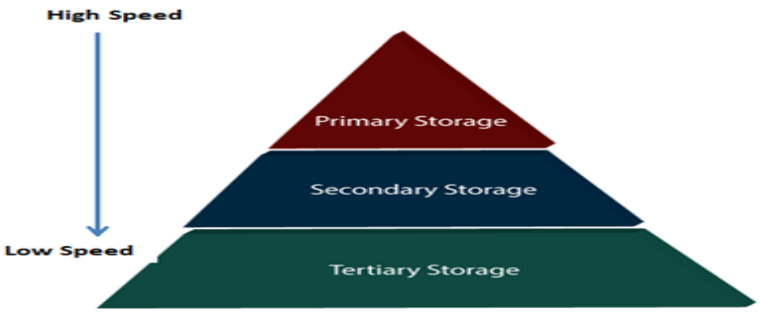
A database system provides an ultimate view of the stored data. However, data in the form of bits, bytes get stored in different storage devices.

Various types of storage devices that are used for accessing and storing data.

Types of Data Storage

For storing the data, there are different types of storage options available. These storage types differ from one another as per the speed and accessibility. There are the following types of storage devices used for storing the data:

- Primary Storage
- Secondary Storage
- Tertiary Storage



Primary Storage − The memory storage that is directly accessible to the CPU comes under this category. CPU's internal memory (registers), fast memory (cache), and main memory (RAM) are directly accessible to the CPU, as they are all placed on the motherboard or CPU chipset. This

storage is typically very small, ultra-fast, and volatile. Primary storage requires continuous power supply in order to maintain its state. In case of a power failure, all its data is lost.

Secondary Storage − Secondary storage devices are used to store data for future use or as backup. Secondary storage includes memory devices that are not a part of the CPU chipset or motherboard, for example, magnetic disks, optical disks (DVD, CD, etc.), hard disks, flash drives, and magnetic tapes.

Tertiary Storage – Tertiary storage is used to store huge volumes of data. Since such storage devices are external to the computer system, they are the slowest in speed. These storage devices are mostly used to take the back up of an entire system. Optical disks and magnetic tapes are widely used as tertiary storage.

References:-1.Database System Concepts 6th Edition Abraham Silberschatz, Henry F. Korth

2. Fundamentals of Database Systems" by Elmsari, Navathe, 5th Edition, Pearson Education (2008)

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UNIT-1 (Queries and Structure of a DBMS)
LECTURE NO: -04

What is a Query?

In simple terms, query as questions or doubts. Query means finding information, asking questions about something, especially in order to express one's doubts about it or to check its validity or accuracy.

A Query is used to traverse over some data may be of small or large quantity to find the needed information.

Data is the facts and figures stored digitally or physically at a location. The data provides us with the base on which we run or perform our query. Books in a library can be considered as data. Information on the web can be considered as data. To look for the desired information in this heap of data is called a query.

We perform many queries in our day to day life. Asking questions is a form of query. Searching info on the web is a form of query. Looking for places on Map is a form of query. In each of these examples, we are looking for the required information in different ways.

A query language is a language which is used to specify our need in the retrieval procedures of information.

A database also contains a lot of data. It stores data in the form of tables. This form of data is used in many places because of its various benefits. A database query refers to the request of data or information from a database. The database consists of tables which contain records of various entities. Thus, a database query is used to query the records of a table. A database query is performed using a database query language which generates data of different types according to function.

A database query can be of two types :

SELECT Query

ACTION Query.

SELECT Query: A select query is used to retrieve data from a single table of a combination of multiple tables.

SQL uses a SELECT statement to select, or extract, specific data based on the given base condition.

ACTION Query: The action query is used to perform operations on the database such as insertion, deletion, modification, alteration, etc.

This query changes the database in one way or another.

Structure of DBMS: -

Database Management System (DBMS) is a software that allows access to data stored in a database and provides an easy and effective method of –

- Defining the information.
- Storing the information.
- Manipulating the information.
- Protecting the information from system crashes or data theft.
- Differentiating access permissions for different users.

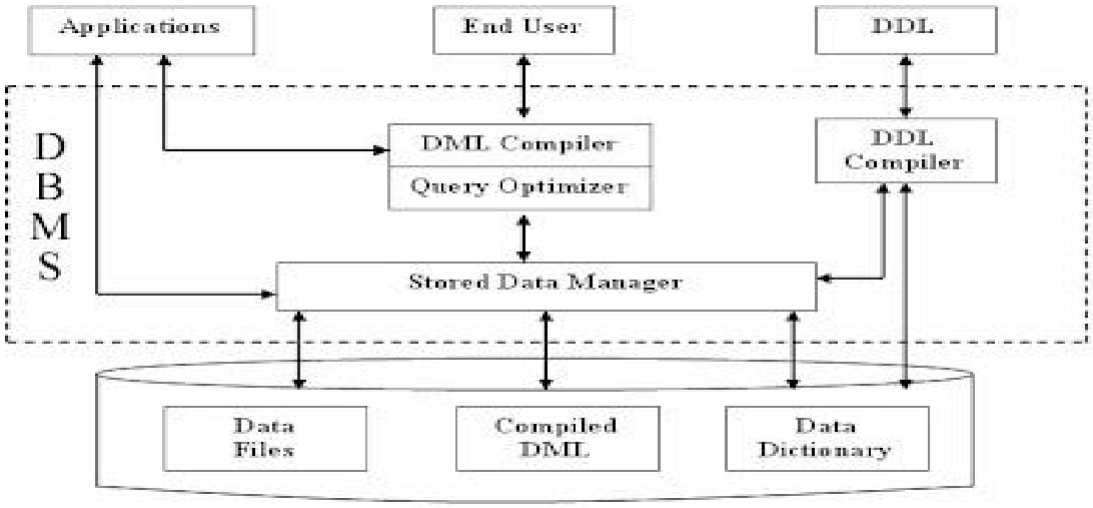


Figure Structure Of DBMS

Applications: – It can be considered as a user-friendly web page where the user enters the requests. Here he simply enters the details that he needs and presses buttons to get the data.

End User: – They are the real users of the database. They can be developers, designers, administrators, or the actual users of the database.

DDL: – Data Definition Language (DDL) is a query fired to create database, schema, tables, mappings, etc in the database. These are the commands used to create objects like tables, indexes in the database for the first time. In other words, they create the structure of the database.

DDL Compiler: – This part of the database is responsible for processing the DDL commands. That means this compiler actually breaks down the command into machine-understandable codes. It is also responsible for storing the metadata information like table name, space used by it, number of columns in it, mapping information, etc.

DML Compiler: – When the user inserts, deletes, updates or retrieves the record from the database, he will be sending requests which he understands by pressing some buttons. But for the database to work/understand the request, it should be broken down to object code.

Query Optimizer: – When a user fires some requests, he is least bothered how it will be fired on the database. He is not all aware of the database or its way of performance. But whatever be the request, it should be efficient enough to fetch, insert, update, or delete the data from the database. The query optimizer decides the best way to execute the user request which is received from the DML compiler.

Storage Manager:

Storage Manager is a program that provides an interface between the data stored in the database and the queries received. It is also known as Database Control System. It maintains the consistency and integrity of the database by applying the constraints and executes the DCL statements. It is responsible for updating, storing, deleting, and retrieving data in the database.

It contains the following components –

Authorization Manager – It ensures role-based access control, i.e,. checks whether the particular person is privileged to perform the requested operation or not.

Integrity Manager – It checks the integrity constraints when the database is modified.

Transaction Manager – It controls concurrent access by performing the operations in a scheduled way that it receives the transaction. Thus, it ensures that the database remains in the consistent state before and after the execution of a transaction.

File Manager – It manages the file space and the data structure used to represent information in the database.

Buffer Manager –It is responsible for cache memory and the transfer of data between the secondary storage and main memory.

Disk Storage -It contains the following components –

Data Files – It stores the data.

Data Dictionary – It contains the information about the structure of any database object. It is the repository of information that governs the metadata.

Indices – It provides faster retrieval of data item.

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References:-1.Database System Concepts 6th Edition Abraham Silberschatz, Henry F. Korth

2.Fundamentals of Database Systems" by Elmsari, Navathe, 5th Edition, Pearson Education (2008)

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UNIT-1 (Entity Relationship Model)
LECTURE NO: -05

Entity Relationship Model: -

ER Model stands for an Entity-Relationship model. It is a high-level data model. This model is used to define the data elements and relationship for a specified system.

It develops a conceptual design for the database. It also develops a very simple and easy to design view of data.

In ER modelling, the database structure is Draw as a diagram called an entity-relationship diagram. the ER model is considered a good option for designing databases.

History of ER models

ER diagrams are visual tools that are helpful to represent the ER model. Peter Chen proposed ER Diagram in 1971 to create a uniform convention that can be used for relational databases and networks. He aimed to use an ER model as a conceptual modelling approach.

Why use ER Diagrams?

- Helps you to define terms related to entity relationship modelling
- Provide a preview of how all your tables should connect, what fields are going to be on each table
- Helps to describe entities, attributes, relationships
- ER diagrams are translatable into relational tables which allows you to build databases quickly
- ER diagrams can be used by database designers as a blueprint for implementing data in specific software applications.

  Examples of entities:

  Person: Employee, Student, Patient

  Place: Store, Building

  Object: Machine, product, and Car

  Event: Sale, Registration, Renewal

  Concept: Account, Course

References:-1.Database System Concepts 6th Edition Abraham Silberschatz, Henry F. Korth

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UNIT-1 (Attributes and Entity Sets)
LECTURE NO: -06

Entity:

An entity is an object or component of data. An entity is represented as rectangle in an ER diagram.

For example: In the following ER diagram we have two entities Student and College and these two entities have many to one relationship as many student's study in a single college. We will read more about relationships later, for now focus on entities.



Weak Entity:

An entity that cannot be uniquely identified by its own attributes and relies on the relationship with other entity is called weak entity. The weak entity is represented by a double rectangle. For example – a bank account cannot be uniquely identified without knowing the bank to which the account belongs, so bank account is a weak entity.

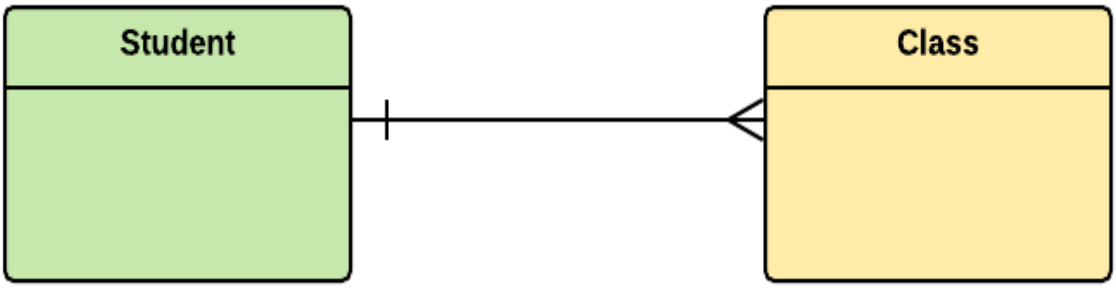
Strong Entity Set	Weak Entity Set
Strong entity set always has a primary key.	It does not have enough attributes to build a primary key.
It is represented by a rectangle symbol.	It is represented by a double rectangle symbol.
It contains a Primary key represented by the underline symbol.	It contains a Partial Key which is represented by a dashed underline symbol.
The member of a strong entity set is called as dominant entity set.	The member of a weak entity set called as a subordinate entity set.
Primary Key is one of its attributes which helps to identify its member.	In a weak entity set, it is a combination of primary key and partial key of the strong entitv set.

Entity set:

Student

An entity set is a group of similar kind of entities. It may contain entities with attribute sharing similar values. Entities are represented by their properties, which also called attributes. All

attributes have their separate values. For example, a student entity may have a name, age, class, as attributes.



ER Diagrams Symbols & Notations

Entity Relationship Diagram Symbols & Notations mainly contains three basic symbols which are rectangle, oval and diamond to represent relationships between elements, entities and attributes. There are some sub-elements which are based on main elements in ERD Diagram. ER Diagram is a visual representation of data that describes how data is related to each other using different ERD Symbols and Notations.

Following are the main components and its symbols in ER Diagrams:

Rectangles: This Entity Relationship Diagram symbol represents entity types

Ellipses: Symbol represent attributes

Diamonds: This symbol represents relationship types

Lines: It links attributes to entity types and entity types with other relationship types

Primary key: attributes are underlined

Double Ellipses: Represent multi-valued attributes

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Attributes and Entity Sets



Attributes

Entities are represented by means of their properties, called attributes. All attributes have values. For example, a student entity may have name, class, and age as attributes.

There exists a domain or range of values that can be assigned to attributes. For example, a student's name cannot be a numeric value. It has to be alphabetic. A student's age cannot be negative, etc.

Types of Attributes

Simple attribute − Simple attributes are atomic values, which cannot be divided further. For example, a student's phone number is an atomic value of 10 digits.

Composite attribute − Composite attributes are made of more than one simple attribute. For example, a student's complete name may have first_name and last_name.

Derived attribute − Derived attributes are the attributes that do not exist in the physical database, but their values are derived from other attributes present in the database. For example, average_salary in a department should not be saved directly in the database, instead it can be derived. For another example, age can be derived from data_of_birth.

Single-value attribute − Single-value attributes contain single value. For example − Social_Security_Number.

Multi-value attribute − Multi-value attributes may contain more than one values. For example, a person can have more than one phone number, email_address, etc.

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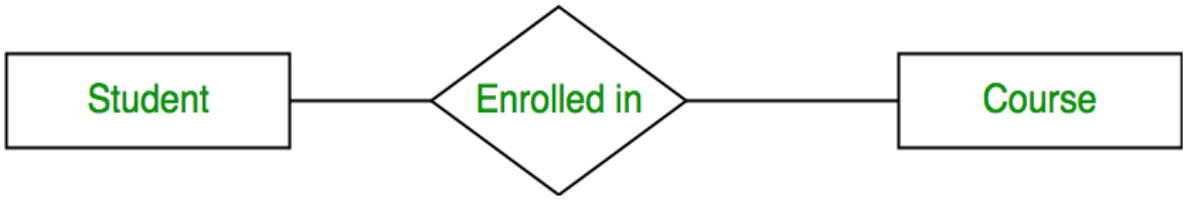
UNIT-1 (Relationship Sets and Features of the ER Model)
LECTURE NO: -07

Relationship

A relationship is represented by diamond shape in ER diagram, it shows the relationship among entities.

Relationship Type and Relationship Set:

A relationship type represents the association between entity types. For example, 'Enrolled in' is a relationship type that exists between entity type Student and Course. In ER diagram, relationship type is represented by a diamond and connecting the entities with lines.

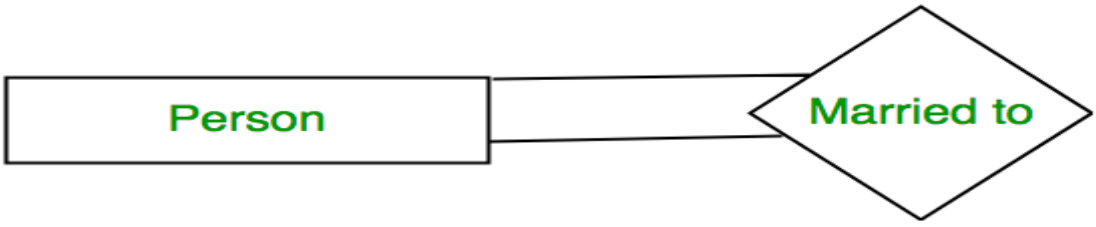


Degree of a relationship set:

The number of different entities sets participating in a relationship set is called as degree of a relationship set.

1. Unary Relationship –

When there is only ONE entity set participating in a relation, the relationship is called as unary relationship. For example, one person is married to only one person.

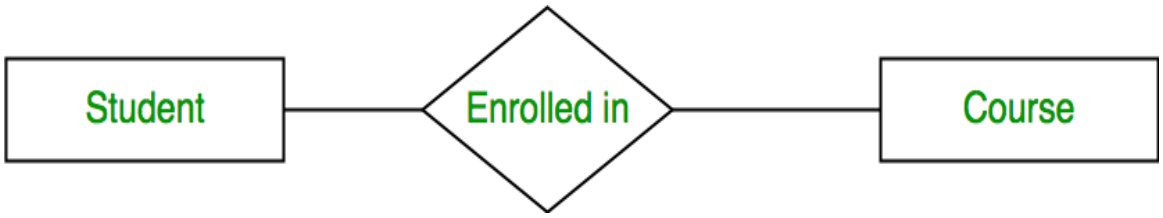


2. Binary Relationship –

When there are TWO entities set participating in a relation, the relationship is called as binary relationship. For example, Student is enrolled in Course.

Notes By: Dr. Rajesh Kumar (Asst. Prof. AI Dept.)

Relationship Sets and Features of the ER Model



Cardinality (Relation Ship): The number of times an entity of an entity set participates in a relationship set is known as cardinality. Cardinality can be of different types:

1. One to One
2. One to Many
3. Many to One
4. Many to Many

1. One-to-One Relationship

When only one instance of an entity is associated with the relationship, then it is known as one to one relationship.

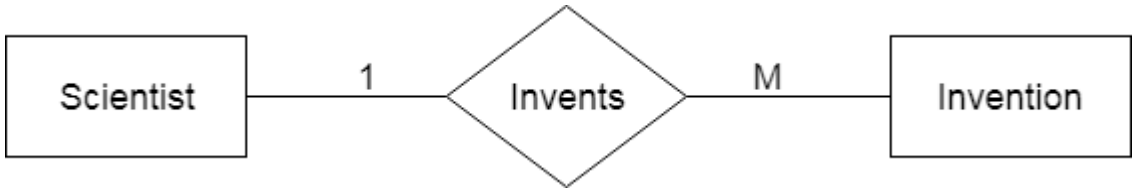
For example, A female can marry to one male, and a male can marry to one female.



2. One-to-many relationship

When only one instance of the entity on the left, and more than one instance of an entity on the right associates with the relationship then this is known as a one-to-many relationship.

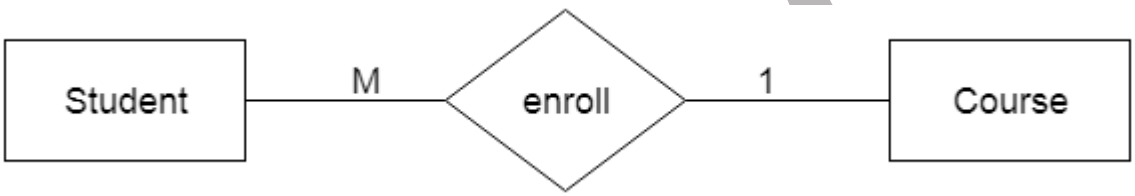
For example, Scientist can invent many inventions, but the invention is done by the only specific scientist.



3. Many-to-one relationship

When more than one instance of the entity on the left, and only one instance of an entity on the right associates with the relationship then it is known as a many-to-one relationship.

For example, Student enrolls for only one course, but a course can have many students.



4. Many-to-many relationship

When more than one instance of the entity on the left, and more than one instance of an entity on the right associates with the relationship then it is known as a many-to-many relationship.

For example, Employee can assign by many projects and project can have many employees.



The components and features of an ER diagram

ER Diagrams are composed of entities, relationships and attributes. They also depict cardinality, which defines relationships in terms of numbers.

Entity

A definable thing—such as a person, object, concept or event—that can have data stored about it. Think of entities as nouns. Examples: a customer, student, car or product. Typically shown as a rectangle.

Relationship Sets and Features of the ER Model

Entity type: A group of definable things, such as students or athletes, whereas the entity would be the specific student or athlete. Other examples: customers, cars or products.

Entity set: Same as an entity type, but defined at a particular point in time, such as students enrolled in a class on the first day. Other examples: Customers who purchased last month, cars currently registered in Florida. A related term is instance, in which the specific person or car would be an instance of the entity set.

Entity categories: Entities are categorized as strong, weak or associative. A strong entity can be defined solely by its own attributes, while a weak entity cannot. An associative entity associates entities (or elements) within an entity set.

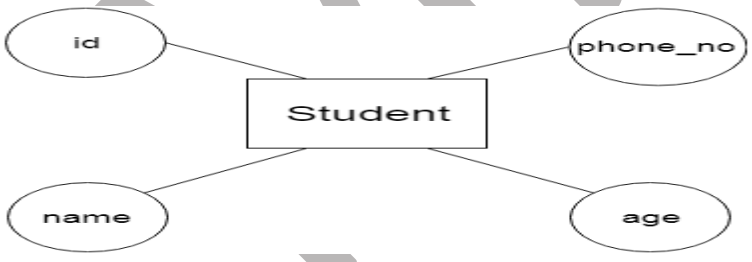
2. Relationship

How entities act upon each other or are associated with each other. Think of relationships as verbs. For example, the named student might register for a course. The two entities would be the student and the course, and the relationship depicted is the act of enrolling, connecting the two entities in that way. Relationships are typically shown as diamonds or labels directly on the connecting lines.

3. Attribute

The attribute is used to describe the property of an entity. Eclipse is used to represent an attribute.

For example, id, age, contact number, name, etc. can be attributes of a student.



References: -

1. Database System Concepts 6th Edition Abraham Silberschatz, Henry F. Korth

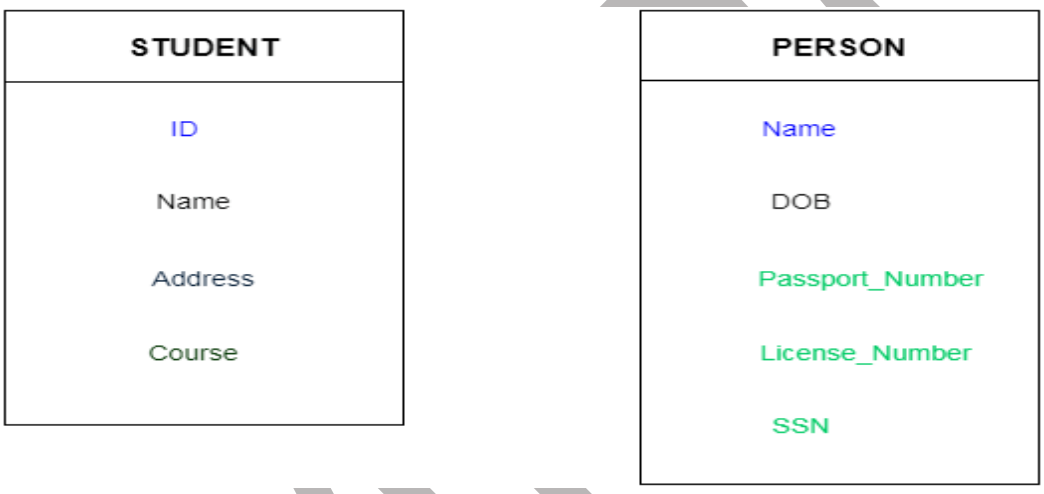
2.Fundamentals of Database Systems" by Elmsari, Navathe, 5th Edition, Pearson Education (2008).

UNIT-1 (Key Constraints)
LECTURE NO: -08

Keys: -

A DBMS key is an attribute or set of an attribute which helps you to identify a row(tuple) in a relation(table). They allow you to find the relation between two tables. Keys help you uniquely identify a row in a table by a combination of one or more columns in that table.

For example: In Student table, ID is used as a key because it is unique for each student. In PERSON table, passport_number, license_number, SSN are keys since they are unique for each person.



Why we need a Key?

Here, are reasons for using Keys in the DBMS system. Keys help you to identify any row of data in a table. In a real-world application, a table could contain thousands of records. Moreover, the records could be duplicated. Keys ensure that you can uniquely identify a table record despite these challenges.

Allows you to establish a relationship between and identify the relation between tables

Help you to enforce identity and integrity in the relationship.

Various Keys in Database Management System: -

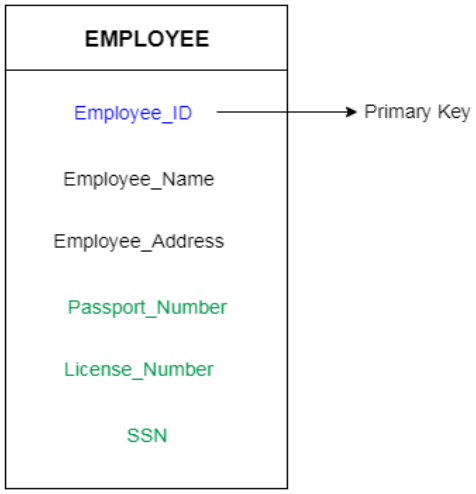
1. Primary Key
2. Candidate Key
3. Super Key

4. Foreign Key

5. Alternative Key

1. Primary key

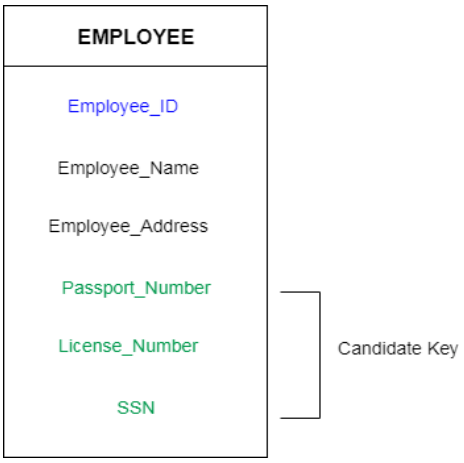
- It is the first key which is used to identify one and only one instance of an entity uniquely. An entity can contain multiple keys as we saw in PERSON table. The key which is most suitable from those lists become a primary key.
- In the EMPLOYEE table, ID can be primary key since it is unique for each employee. In the EMPLOYEE table, we can even select License_Number and Passport_Number as primary key since they are also unique.
- For each entity, selection of the primary key is based on requirement and developers.



2. Candidate key

- A candidate key is an attribute or set of an attribute which can uniquely identify a tuple.
- The remaining attributes except for primary key are considered as a candidate key. The candidate keys are as strong as the primary key.

For example: In the EMPLOYEE table, id is best suited for the primary key. Rest of the attributes like SSN, Passport_Number, and License_Number, etc. are considered as a candidate key.



3. Super Key

Super key is a set of an attribute which can uniquely identify a tuple. Super key is a superset of a candidate key.

For example: In the above EMPLOYEE table, for(EMPLOEE_ID, EMPLOYEE_NAME) the name of two employees can be the same, but their EMPLYEE_ID can't be the same. Hence, this combination can also be a key.

4. Foreign key

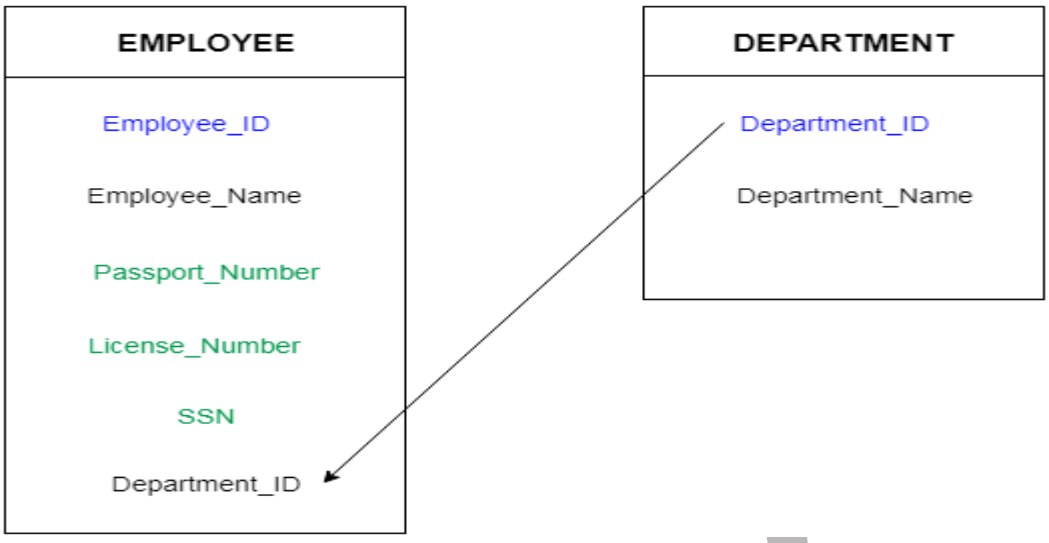
Foreign keys are the column of the table which is used to point to the primary key of another table.

In a company, every employee works in a specific department, and employee and department are two different entities. So we can't store the information of the department in the employee table. That's why we link these two tables through the primary key of one table.

We add the primary key of the DEPARTMENT table, Department_Id as a new attribute in the EMPLOYEE table.

Now in the EMPLOYEE table, Department_Id is the foreign key, and both the tables are related.

Key Constraints



Difference Between Primary key & Foreign key

Following is the main difference between primary key and foreign key:

Primary Key	Foreign Key
Helps you to uniquely identify a record in the table.	It is a field in the table that is the primary key of another table.
Primary Key never accept null values.	A foreign key may accept multiple null values.
Primary key is a clustered index and data in the DBMS table are physically organized in the sequence of the clustered index.	A foreign key cannot automatically create an index, clustered or non-clustered. However, you can manually create an index on the foreign key.
You can have the single Primary key in a table.	You can have multiple foreign keys in a table.

References: -

1. Database System Concepts 6th Edition Abraham Silberschatz, Henry F. Korth

2.Fundamentals of Database Systems" by Elmsari, Navathe, 5th Edition, Pearson Education (2008).

3. C.J. Date: Data Base Design, Addison Wesley

UNIT-1 (Participation Constraints and Class Hierarchies, Aggregation and Conceptual Data Base)
LECTURE NO: -09

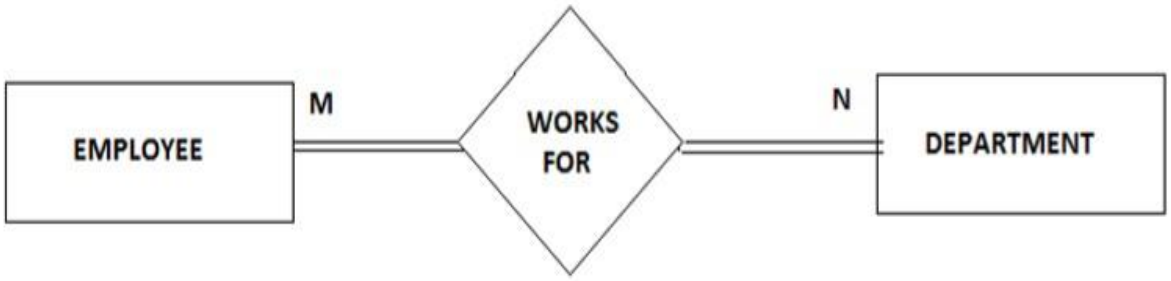
In a Relationship, Participation constraint specifies the existence of an entity when it is related to another entity in a relationship type. It is also called minimum cardinality constraint.

This constraint specifies the number of instances of an entity that can participate in a relationship type.

There are two types of Participation constraint −

Total Participation

Each entity in the entity set is involved in at least one relationship in a relationship set i.e. the number of relationships in every entity is involved is greater than 0.



Consider two entities Employee and Department related via Works_For relationship. Now, every Employee works in at least one department therefore an Employee entity exist if it has at least one Works_For relationship with Department entity. Thus, the participation of Employee in Works_For is total relationship.

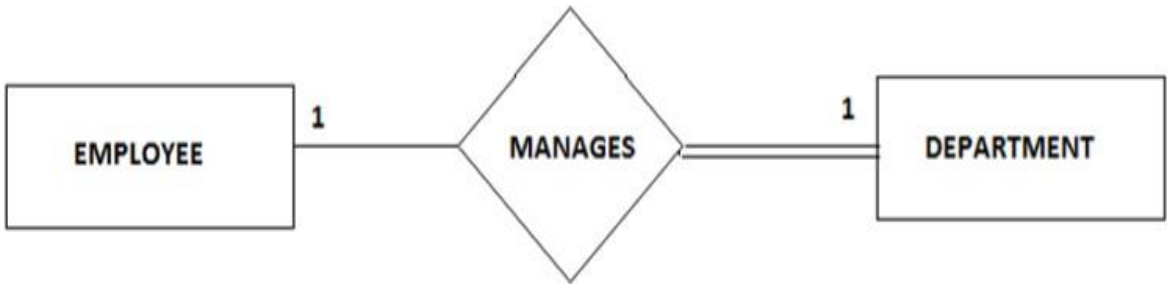
Total Participation is represented by double line in ER diagram.

Partial Participation

Each entity in entity set may or may not occur in at least one relationship in a relationship set.

For example: Consider two entities Employee and Department and they are related to each other via Manages relationship. An Employee must manage a Department, he or she could be the head of the department. But not every Employee in the company manages the department. So, participation of employee in the Manages relationship type is partial i.e. only a particular set of Employees will manage the Department but not all.

Participation Constraints and Class Hierarchies, Aggregation and Conceptual Data Base



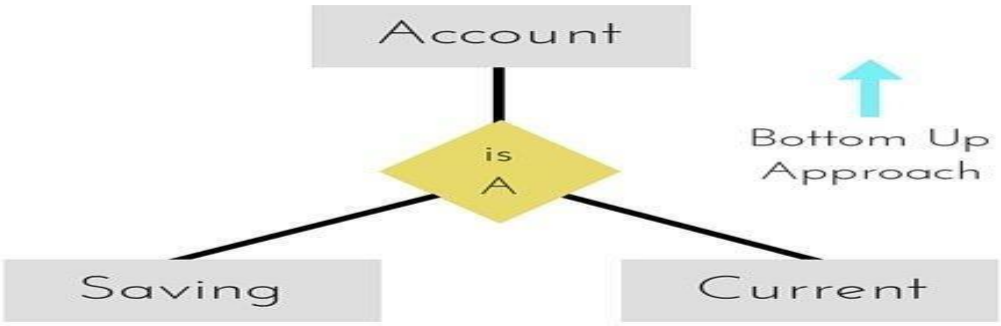
Class Hierarchies: -

The ER Model has the power of expressing database entities in a conceptual hierarchical manner. As the hierarchy goes up, it generalizes the view of entities, and as we go deep in the hierarchy, it gives us the detail of every entity included.

Going up in this structure is called generalization, where entities are clubbed together to represent a more generalized view. For example, a particular student named Mira can be generalized along with all the students. The entity shall be a student, and further, the student is a person. The reverse is called specialization where a person is a student, and that student is Mira.

Generalization

Generalization is a bottom-up approach in which two lower level entities combine to form a higher-level entity. In generalization, the higher-level entity can also combine with other lower level entities to make further higher-level entity. It's more like Superclass and Subclass system, but the only difference is the approach, which is bottom-up. Hence, entities are combined to form a more generalised entity, in other words, sub-classes are combined to form a super-class.

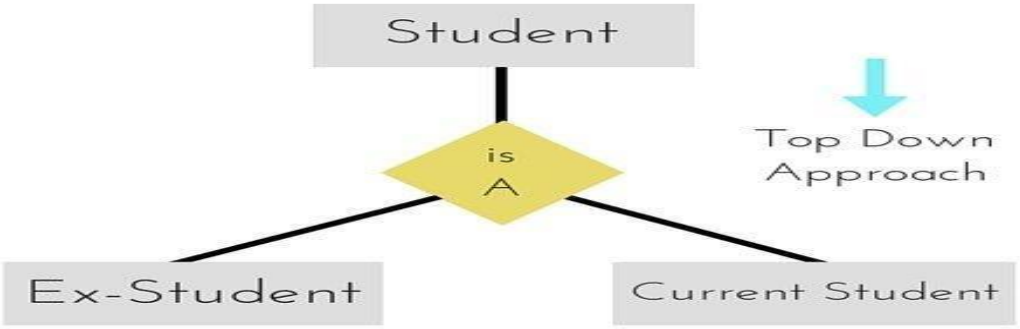


For example, Saving and Current account types entities can be generalised and an entity with name Account can be created, which covers both.

Specialization

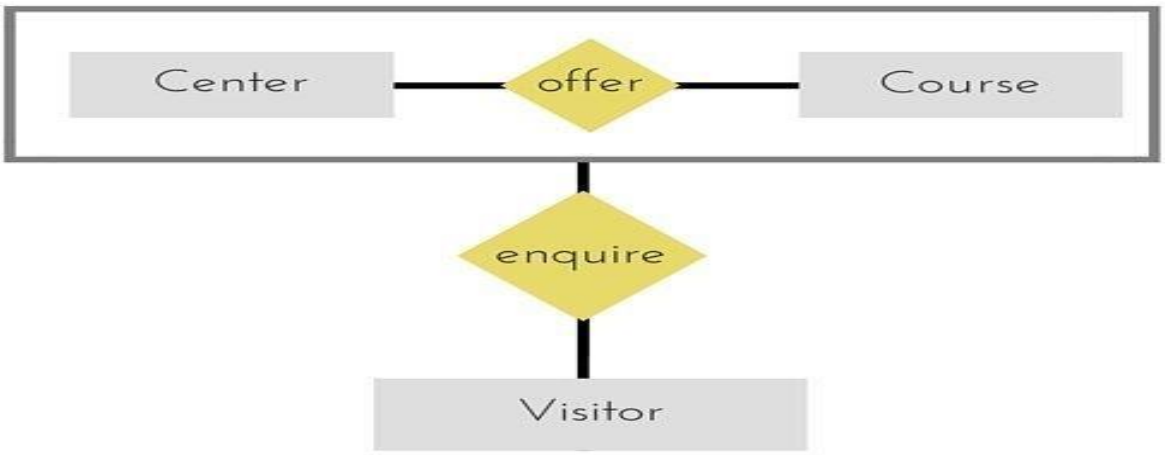
Specialization is the opposite of generalization. In specialization, a group of entities is divided into sub-groups based on their characteristics.

Specialization is opposite to Generalization. It is a top-down approach in which one higher level entity can be broken down into two lower level entity. In specialization, a higher-level entity may not have any lower-level entity sets, it's possible.



Aggregation

Aggregation is a process when relation between two entities is treated as a single entity.



In the diagram above, the relationship between Centre and Course together, is acting as an Entity, which is in relationship with another entity Visitor. Now in real world, if a Visitor or a Student visits a Coaching Centre, he/she will never enquire about the centre only or just about the course, rather he/she will ask enquire about both.

References: -1. Database System Concepts 6th Edition Abraham Silberschatz, Henry F. Korth

UNIT-1 (ER Model Entity vs Attribute and Entity vs Relationship)
LECTURE NO: -10

The ER model defines the conceptual view of a database. It works around real-world entities and the associations among them. At view level, the ER model is considered a good option for designing databases.

Entity

An entity can be a real-world object, either animate or inanimate, that can be easily identifiable. For example, in a school database, students, teachers, classes, and courses offered can be considered as entities. All these entities have some attributes or properties that give them their identity.

An entity set is a collection of similar types of entities. An entity set may contain entities with attribute sharing similar values. For example, a Students set may contain all the students of a school; likewise, a Teachers set may contain all the teachers of a school from all faculties. Entity sets need not be disjoint.

Attributes

Entities are represented by means of their properties, called attributes. All attributes have values. For example, a student entity may have name, class, and age as attributes.

There exists a domain or range of values that can be assigned to attributes. For example, a student's name cannot be a numeric value. It has to be alphabetic. A student's age cannot be negative, etc.

Types of Attributes

Simple attribute − Simple attributes are atomic values, which cannot be divided further. For example, a student's phone number is an atomic value of 10 digits.

Composite attribute − Composite attributes are made of more than one simple attribute. For example, a student's complete name may have first_name and last_name.

Derived attribute − Derived attributes are the attributes that do not exist in the physical database, but their values are derived from other attributes present in the database. For example, average_salary in a department should not be saved directly in the database, instead it can be derived. For another example, age can be derived from data_of_birth.

Notes By: Dr. Rajesh Kumar (Asst. Prof. AI Dept.)

ER Model Entity vs Attribute and Entity vs Relationship

Single-value attribute –Single-value attributes contain single value. For example – Social_Security_Number.

Multi-value attribute – Multi-value attributes may contain more than one values. For example, a person can have more than one phone number, email_address, etc.

Basis of Differentiation	Entity	Attribute
Basic	An entity is a distinguishable real-world object that exist.	An attribute describes the elementary feature of an entity.
Relational Data Model	In the relational data model, an entity is represented as a record in an entity set.	In the relational data model, a field represents an attribute.
Table	A row in a database table is an entity.	A column header of a database table is an attribute.
Grouping	Entities of similar type are placed in one entity set/table.	Group of attributes that easily identifies a tuple forms a key.
Refinement	An entity can be a tangible entity or an intangible entity.	Each attribute has a certain domain and it may be involved in forming a key.

ER Model Entity vs Attribute and Entity vs Relationship

Basis of Differentiation	Entity	Attribute
Example	A student tuple in a student table in the database represents an entity.	Name, roll number, Id, Address, these fields of student table presents the attribute of student entity.

Difference Between Entity and Relationship in DBMS

The main difference between entity and relationship in DBMS is that the entity is a real-world object while the relationship is an association between the entities. Also, in the ER diagram, a rectangle represents an entity while a rhombus or diamond represents a relationship.

A Database Management System (DBMS) is a software program that stores, retrieves and manipulates data in the databases. A DBMS contains multiple databases, and each database consists of multiple tables. The tables are related to each other using relationships. DBMS provides multiple advantages. The user can change and retrieve data in DBMS using Structured Query Language (SQL). An entity is a table in DBMS, and it represents a real-world object. These entities are connected to each other using relationships.

Difference Between ER and Relationship in DBMS

Comparison Basis	ER Model	Relational Model
Basic	It's used to describe a set of objects known as entities, as well as the relationships between them.	It's used to represent a collection of tables as well as the relationships between them.
Type	It is a high-level or conceptual model.	It is the implementation or representational model.
Components	It represents components as Entity, Entity Type, and Entity Set.	It represents components as domain, attributes, and tuples.

ER Model Entity vs Attribute and Entity vs Relationship

Used By	This model is helpful for those people who don't have any knowledge about how data is implemented.	This model is mostly famous among programmers.
Relationship	It is easy to understand the relationship between entities.	Compared to ER model, it is easier to derive the relation between tables in the Relational model.
Mapping	This model explains how to map Cardinalities. The uniqueness of data values in a row is referred to as cardinality.	This model does not describe mapping cardinalinality.

References: -1. Database System Concepts 6th Edition Abraham Silberschatz, Henry F. Korth

2. Fundamentals of Database Systems" by Elmsari, Navathe, 5th Edition, Pearson Education (2008)

UNIT-1 (Binary and Ternary Relationship and Aggregation vs ternary) LECTURE NO: -11

What are Relationships?

A relationship in a DBMS, is primarily the way two or more data sets are linked. This is so true for Relational Database Management Systems. One dataset may be then termed as the Foreign key and the ones linked to it may be termed as the Primary Key. There may be multiple Foreign and Primary keys linked to each other.

Relationships allow the datasets to share and store the data in separate tables. They also help link disparate data with each other.

The most common types of relationships are:

- Unary (one entity is involved in the relationship).
- Binary (two entities are involved in the relationship).
- Ternary (three entities are involved in the relationship)
- N-ary (n entities involved in the relationship)

A unary relationship is when both participants in the relationship are the same entity. For Example: Subjects may be prerequisites for other subjects. A ternary relationship is when three entities participate in the relationship.

Example of unary relationship

For example, a person has only one passport and only one passport is given to only one person and hence unary relationship is observed

A Binary Relationship is the relationship between two different Entities i.e. it is a relationship of role group of one entity with the role group of another entity.

Example of binary relationship

For example, a teacher teaches a subject here 2 entities are teacher and subject for the relationship teacher teaches subject

Ternary Relationship: a ternary relationship is a relationship of degree three. That is, a relationship that contains three participating entities. The cardinality constraint of an entity in a ternary relationship is defined by a pair of two entity instances associated with the other single entity instance.

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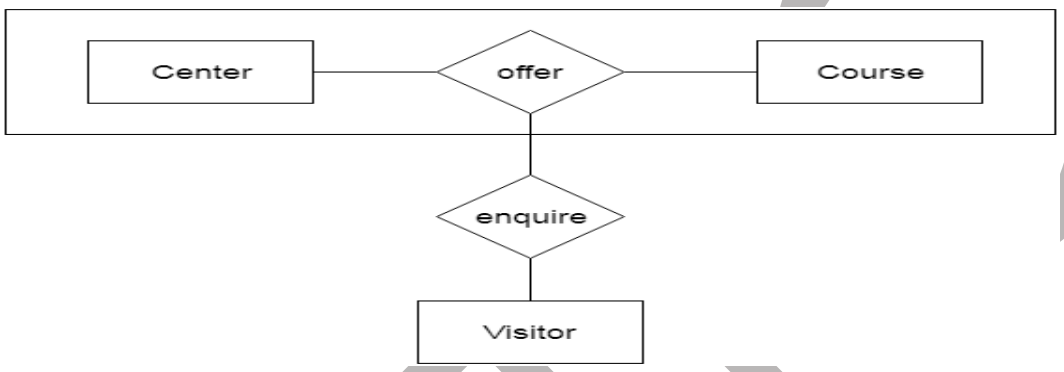
Binary and Ternary Relationship and Aggregation vs ternary

Example of Ternary Relationship

In the real world, a patient goes to a doctor and doctor prescribes the medicine to the patient, three entities Doctor, patient and medicine are involved in the relationship "prescribes".

Aggregation in DBMS

Aggregation refers to the process by which entities are combined to form a single meaningful entity. The specific entities are combined because they do not make sense on their own. To establish a single entity, aggregation creates a relationship that combines these entities. The resulting entity makes sense because it enables the system to function well.



When using data in the form of numerical values, the following operations can be used to perform DBMS aggregation:

Average (AVG): This function provides the mean or average of the data values.

Sum: This provides a total value after the data values have been added.

Count: This provides the number of records.

Maximum (Max): This function provides the maximum value of a given set of data.

Minimum (Min): This provides the minimum value of a given set of data.

References: -1. Database System Concepts 6th Edition Abraham Silberschatz, Henry F. Korth

2. Fundamentals of Database Systems" by Elmsari, Navathe, 5th Edition, Pearson Education (2008)